

THE PROBLEM

Cannot handle heterogeneous devices in federated learning



DATA PRIVACY

Regulatory Barriers

- GDPR / HIPAA
- Medical data silos

Grok Leak (2025): 370k+ chats



HARDWARE GAP

High-Tier

Rank 32

48GB VRAM

Full capacity

Edge-Tier

Rank 4

8GB VRAM

Limited



CURRENT FL

Forces Same Rank

- Excludes high-capacity

Bottleneck: $r_{\min}$



Research Question

How to enable privacy-preserving federated learning where each device operates at its optimal rank?

Our Solution: Subspace Projection Aggregation (SPA)

1

Heterogeneous

Each device optimal rank

2

SVD Distillation

High→low rank projection

3

Decentralized

Privacy preserved